Customer Value & Promotion Targeting for an Online Retailer

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Abstract

This study develops a Business Analytics framework for customer segmentation, purchasing prediction, and promotion targeting using the UCI Online Retail II dataset. RFM segmentation and cohort analysis characterize customer behavior, while Logistic Regression, Linear Regression, and Gradient Boosting are evaluated using chronological data splits. On the final test set of 5,281 customers, Logistic Regression achieves an Average Precision of 0.7502 for repeat-purchase prediction, and Linear Regression achieves an MAE of £395.69 for 90-day gross-sales forecasting. These predictions support a budget-constrained promotion model incorporating discounts, assumed sales growth, margins, and contact costs. Under three reference scenarios, the proposed policy recommends no promotional contacts because every candidate contact has a negative simulated incremental margin. Sensitivity analysis identifies potentially profitable alternative scenarios. The findings highlight the importance of economically informed targeting, while acknowledging that promotional responses are assumed rather than causally estimated.

1 Introduction and Business Understanding

1.1 Background and Business Problem

Customer retention and promotional targeting are central challenges in online retail, where businesses must balance customer engagement with marketing expenditure and profitability. Historical transaction data provide valuable insights into purchasing frequency, recency, and monetary value, enabling retailers to segment customers and anticipate future behavior. However, customers with high predicted spending are not necessarily the most profitable recipients of promotional offers. Discounts may reduce margins on purchases that would have occurred without an incentive, making customer value and promotional profitability fundamentally different concepts.

This study addresses the business question: *Which customers should the retailer prioritize for retention or promotion to increase expected future margin without over-discounting?* From the perspective of a marketing or customer relationship management (CRM) decision-maker, the objective is to determine whether a promotion should be offered, which customers should receive it, and how a limited contact budget should be allocated. Rather than maximizing campaign reach or predicted customer spending, the proposed framework evaluates promotional decisions according to their simulated net incremental margin, allowing the retailer to abstain when no economically favorable opportunities exist.

Addressing this problem requires integrating descriptive, predictive, and prescriptive analytics. Customer segmentation characterizes historical purchasing patterns, predictive models estimate future purchasing outcomes, and promotion optimization translates these estimates into financial decisions under explicit assumptions. Importantly, the available data do not identify randomized promotional

treatments or corresponding control outcomes. Consequently, this study evaluates scenario-based economic decisions rather than claiming to estimate causal promotional effects.

1.2 Research Objectives and Scope

The primary objective is to develop a reproducible Business Analytics framework that connects customer behavior analysis with predictive modeling and economically informed promotion targeting. Specifically, the study examines how transaction histories can characterize customer segments, how accurately future purchasing behavior and spending can be predicted, and whether these predictions can support profitable promotional decisions under budget constraints. It further investigates the sensitivity of those decisions to economic assumptions and evaluates how the results can be communicated through an interactive dashboard.

The analysis uses the UCI Online Retail II dataset, which contains 1,067,371 transaction-line records from a UK-based non-store retailer [1]. The unit of analysis is a customer–decision-date observation, constructed from historical transactions available before each decision date. Two predictive outcomes are considered over a 90-day horizon: repeat purchase and future positive gross sales. The final temporal-test population contains 5,281 eligible customers at the decision date of 10 September 2011. The monetary target represents positive gross sales rather than returns-adjusted revenue, profit, or lifetime customer value.

The analytical framework combines rule-based Recency–Frequency–Monetary (RFM) segmentation, cohort analysis, and supervised prediction using interpretable baseline models and Gradient Boosting alternatives. Chronological training, validation, and test periods are employed to assess predictive generalization while preventing decision-time information leakage. Classification performance is evaluated through Average Precision, ROC-AUC, and calibration diagnostics, while regression performance is measured primarily through MAE and RMSE. The promotion component compares No Promotion, Non-targeted Promotion, RFM-based Targeting, and a budget-constrained policy that selects customers according to positive simulated net incremental contribution.

The business analysis considers explicit assumptions regarding discount rates, proportional sales-volume growth, gross margins, and contact costs. Break-even calculations, sensitivity analysis, and robustness checks are used to assess the stability of the resulting recommendations. An interactive Streamlit dashboard provides scenario exploration and customer-level decision support. The scope is retrospective: promotional responses are assumed rather than experimentally estimated, and simulated financial outcomes are not interpreted as realized campaign profits.

1.3 Related Work

RFM analysis is an established approach to understanding customer behavior through recency, frequency, and monetary value. Fader et al. [2] demonstrate how RFM information can contribute to forward-looking customer valuation, while Chen et al. [3] investigate RFM-based segmentation in online retail using data-mining techniques. Building on this literature, the present study uses RFM to characterize customer heterogeneity and establish an interpretable targeting benchmark. Unlike customer lifetime value models, however, its monetary prediction task is restricted to a 90-day future gross-sales horizon.

Predictive customer analytics and promotion-effect estimation address related but distinct problems. While conventional classification and regression models estimate future purchasing outcomes from observed characteristics, uplift modeling seeks to estimate the incremental effect of an intervention. Gutierrez and Gérardy [4] review this causal modeling perspective and its relevance to treatment targeting. Because the available retail dataset lacks the treatment information necessary for reliable causal identification, this study does not implement uplift modeling. Instead, it evaluates promotional decisions through a transparent economic simulation with explicitly assumed sales-volume responses.

The overall analytical process follows the business-oriented principles of CRISP-DM [5], connecting business understanding, data preparation, modeling, and evaluation. The present study emphasizes the subsequent decision layer by integrating predictive customer-value estimates with promotion economics, budget-constrained selection, and sensitivity analysis. This positioning distinguishes the framework from approaches that evaluate predictive accuracy or customer rankings without explicitly considering the financial consequences of marketing actions.

1.4 Contributions and Paper Organization

The contribution of this study is an integrated and reproducible decision-support framework rather than a new machine-learning algorithm. It combines temporally evaluated customer predictions with explicit promotion economics, compares conventional and economically informed targeting policies, and examines the conditions under which promotional decisions change. A key feature is the ability to recommend no promotional contact when all available actions have nonpositive simulated contributions. The framework is further supported by robustness analyses and an interactive dashboard that makes its assumptions and financial implications accessible to decision-makers.

The remainder of the paper is organized as follows. Sections 2–4 describe the dataset, analytical preparation, and customer segmentation results. Sections 5–6 present predictive modeling and evaluation. Sections 7–8 examine promotion targeting, simulated business impact, sensitivity, and robustness. Section 9 introduces the decision-support dashboard and managerial implications, while Section 10 concludes with the study's limitations and future research directions.

2 Data Understanding

2.1 Dataset Description and Data Quality

This study uses the UCI Online Retail II dataset, containing 1,067,371 transaction-line records from a UK-based non-store retailer between December 2009 and December 2011 [1]. The source workbook comprises two worksheets, covering 2009–2010 and 2010–2011, which are combined for analysis. The project repository records 26 October 2026 as the data acquisition date. Each record represents an invoice line rather than an individual transaction or customer, and monetary values are expressed in pounds sterling (GBP).

The dataset contains eight attributes: `Invoice`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `Price`, `Customer ID`, and `Country`. Invoice and customer identifiers support transaction aggregation, while product codes, quantities, prices, and timestamps describe purchasing behavior. The distinction between invoice lines and distinct invoices is important because a single purchase may contain multiple products. The complete field descriptions and data provenance are documented in the project data dictionary.

Data quality assessment identified two major issues. First, 243,007 records (22.77%) have missing customer identifiers, preventing reliable customer-level aggregation; these records are excluded from customer-level modeling. Second, 34,335 records (3.22%) are exact duplicates beyond their first occurrences. Since identical invoice lines cannot be conclusively classified as erroneous duplicates, the primary analysis retains them, while a separate robustness experiment evaluates their removal. These counts describe potentially overlapping conditions in the original dataset and should not be interpreted as mutually exclusive cleaning categories.

The data also contain cancellations and nonpositive transaction quantities or prices. Valid purchasing activity is defined using positive-quantity, positive-price, non-cancelled transactions, as detailed in Section 3. Returns are not matched to their original purchases to reconstruct refunded amounts. Consequently, the monetary measures used in this study represent positive gross sales rather than returns-adjusted net revenue or realized profit.

2.2 Customer Population and Purchasing Patterns

Descriptive analysis focuses on the final evaluation population of 5,281 eligible customers at the decision date of 10 September 2011. Each customer has at least one valid historical purchase before that date. The subsequent 90-day purchasing outcomes are examined retrospectively to characterize the evaluation population; they were not available when customer-level predictions would have been generated.

Within the future observation horizon, 2,290 customers (43.36%) make at least one valid purchase, while 2,991 (56.64%) record no positive purchasing activity. The distribution of future gross sales is strongly right-skewed, with a mean of £543.11 and a median of zero. Table 1 summarizes historical and future spending for the same customer population. These quantities cover different observation

periods and should therefore not be interpreted as directly comparable monthly or annual spending rates.

Table 1: Customer-level monetary distributions at the final evaluation snapshot ($N = 5{,}281$). Values are in GBP.

Statistic	Historical gross sales	Future 90-day gross sales
Mean	2,705.44	543.11
Median	804.02	0.00
75th percentile	2,118.75	435.47
95th percentile	8,729.38	1,881.86
Maximum	484,615.10	123,737.11

Figure 1 illustrates the distribution of positive future gross sales using a logarithmic horizontal scale. Customers with zero future sales are reported separately because they cannot be represented on that scale. The distribution exhibits a pronounced upper tail: the 53 customers with the highest observed future gross sales, approximately 1% of the evaluation population, account for 37.58% of total future gross sales. Similarly, the top 53 customers ranked by historical monetary value account for 30.83% of historical gross sales, although the two rankings do not necessarily contain the same individuals.

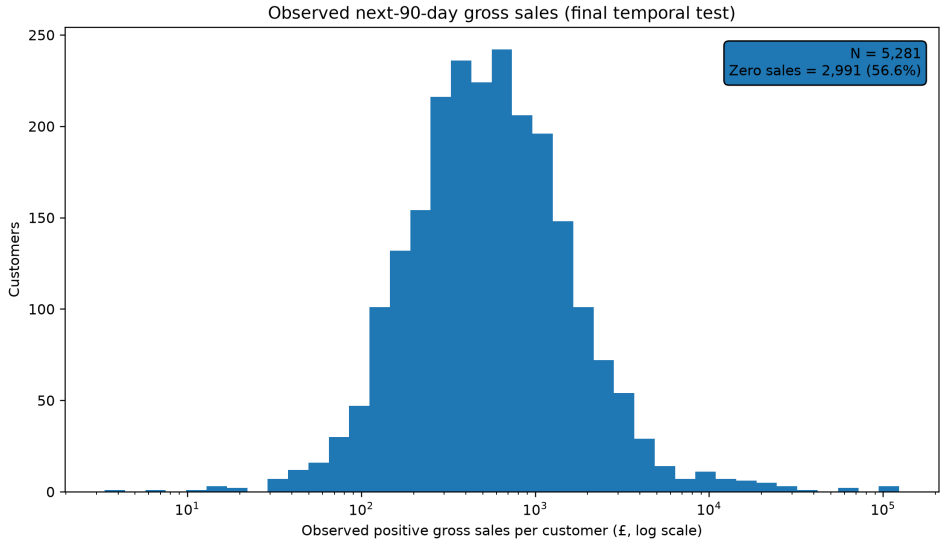


Figure 1: Distribution of positive 90-day future gross sales on a logarithmic scale. The 2,991 customers with zero observed future sales are excluded from the histogram and reported separately.

These characteristics reveal two important modeling challenges: a substantial proportion of customers generate no purchases within the prediction horizon, while a small group contributes disproportionately to total sales. This motivates the complementary use of repeat-purchase classification and monetary forecasting, with particular attention to errors among high-spending customers. Customer heterogeneity is explored further through RFM segmentation and cohort analysis in Section 4.

3 Analytical Design and Data Preparation

3.1 Analytical Framework

The study adopts a decision-centered analytical framework that connects historical customer behavior with predictive modeling and promotion targeting. As illustrated in Figure 2, the workflow begins with transaction validation and preparation, followed by the construction of customer-level observations, behavioral segmentation, and prediction of future purchasing outcomes. Customer-value forecasts are subsequently combined with explicit promotion assumptions to evaluate budget-constrained marketing decisions.

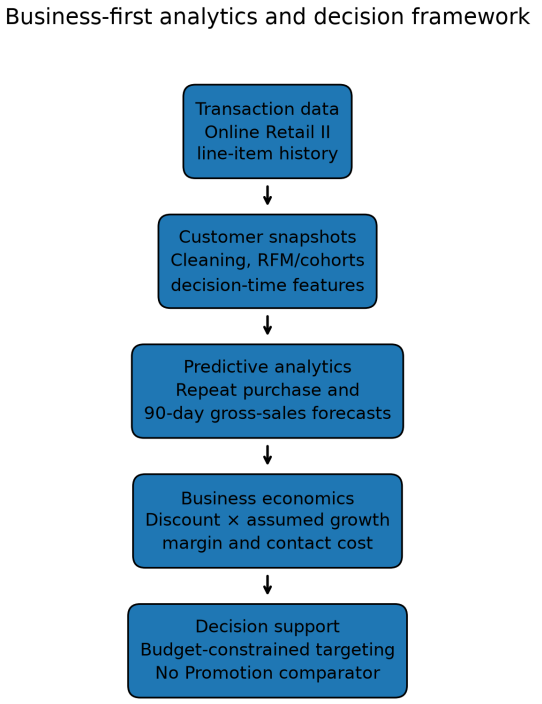


Figure 2: Overview of the analytical workflow, from transaction preparation and customer-level prediction to scenario-based promotion decisions.

A central design principle is to distinguish observed, predicted, assumed, and simulated quantities. Historical transaction records provide observed behavioral features, while predictive models estimate future purchasing outcomes. Discount rates, sales-volume growth, margins, and contact costs are specified as economic assumptions rather than estimated promotional treatment effects. The decision model then combines these inputs to calculate simulated incremental contributions. This separation preserves the distinction between forecasting customer behavior and evaluating hypothetical marketing interventions.

3.2 Transaction Preparation

The analytical pipeline combines the two worksheets of the Online Retail II dataset and validates their transaction fields before constructing customer-level observations. Since the original records represent invoice line items, customer purchasing behavior must be aggregated using customer identifiers and distinct purchase invoices. Records with missing customer identifiers are excluded because they cannot be reliably assigned to individual customers, as discussed in Section 2.

A valid purchase is defined as a non-cancelled transaction line with strictly positive quantity and unit price. Invoices beginning with the character `C` are identified as cancellations and excluded from positive-sales calculations. The remaining records are filtered for nonpositive quantities and prices. For each valid line j, gross sales are calculated as $s_j = q_j p_j$, where q_j and p_j denote quantity and unit price, respectively. The same purchase-validity rules are applied when constructing historical features and future outcomes.

Exact duplicate rows are retained in the primary analysis because the dataset lacks a reliable unique transaction-line identifier, making it difficult to distinguish recording errors from legitimate repeated line items. Their removal is evaluated separately as a robustness check in Section 8.4. Cancellation records are also preserved temporarily during feature construction to measure historical cancellation behavior, but they are not matched to original purchases to calculate refunds. Consequently, monetary

aggregates represent positive gross sales rather than returns-adjusted net revenue. No external datasets are joined to construct the predictive features.

3.3 Temporal Customer Snapshots and Outcome Construction

The unit of analysis is a customer–decision-date snapshot, denoted by (i, T), where i identifies a customer and T represents the date at which a prediction or marketing decision would be made. A customer is eligible for a snapshot if at least one valid purchase is recorded before T. Historical observation windows expand from 1 December 2009 to each decision date, allowing the same customer to appear at multiple origins with updated behavioral characteristics.

To prevent temporal leakage, features are constructed exclusively from transactions satisfying $t_j < T$, where t_j is the transaction timestamp. Outcomes are calculated from the subsequent 90-day interval, $[T, T + 90 \text{ days})$. The half-open intervals ensure that transactions at the decision boundary cannot simultaneously contribute to historical predictors and future outcomes.

Two supervised learning targets are constructed from each snapshot. The repeat-purchase target is a binary indicator of whether the customer records at least one valid positive purchase during the next 90 days. The future gross-sales target is the sum of positive line-item sales over the same period, with a value of zero for customers without a qualifying purchase. These targets support classification and regression, respectively, but neither represents a causal estimate of promotional response or customer lifetime profit.

The main experiment uses two training origins, one validation origin, and one final-test origin. All training outcomes are fully observable before the corresponding evaluation decision date, and final-test labels are excluded from model fitting and selection. The detailed schedule, preprocessing protocol, and model configurations are presented in Section 5.

3.4 Behavioral Feature Engineering

Seven numerical predictors are constructed from each customer's pre-decision transaction history. These variables summarize purchasing recency, frequency, monetary value, product diversity, tenure, and cancellation behavior. Table 2 provides their operational definitions.

Table 2: Customer-level behavioral features constructed using pre-decision transactions.

Feature	Definition
`recency_days`	Days since the most recent valid purchase.
`frequency`	Number of distinct valid purchase invoices.
`monetary`	Total historical positive gross sales (GBP).
`avg_order_value`	Historical gross sales divided by purchase frequency.
`n_distinct_products`	Number of distinct purchased product codes.
`tenure_days`	Days since the first observed valid purchase.
`return_rate`	Share of distinct observation-period invoices identified as cancellations.

The cancellation-related feature is calculated from identifiable observation-period invoices before positive-purchase filtering. It measures the proportion of distinct invoices identified as cancellations, rather than the monetary value or number of returned products. Eligible customers have defined purchase histories, and missing cancellation shares after aggregation are assigned zero.

Customer identifiers and countries are retained for traceability and descriptive analysis but are not included in the predictive feature vector. Similarly, RFM segment labels, cohort assignments, future outcomes, and promotion recommendations are excluded from model inputs. These restrictions ensure that prediction features represent information available at the decision date. Model-specific scaling and estimator configurations are described in Section 5.

3.5 RFM Segmentation and Cohort Definition

In addition to predictive features, Recency–Frequency–Monetary (RFM) scores are constructed to characterize customer heterogeneity and provide an interpretable targeting benchmark. Within each

decision-date snapshot, recency, frequency, and monetary values are converted into quartile scores from 1 to 4. More recent purchasing receives a higher recency score, while greater frequency and spending receive higher scores in their respective dimensions. The three scores are summed to form an RFM score between 3 and 12.

Customers are assigned to four rule-based segments: Champions (scores 10–12), Loyal (8–9), At-risk (6–7), and Lost (3–5). Because quartiles are calculated within each snapshot, segment membership reflects relative purchasing behavior rather than fixed monetary or activity thresholds. The labels are descriptive and do not establish permanent churn status or sensitivity to promotional offers.

Cohorts are defined by the calendar month of each customer's first observed valid purchase. Monthly cohort activity measures the proportion of customers purchasing in each subsequent observed month relative to the original cohort size. Since the dataset has a finite observation period, the first observed purchase may not represent actual customer acquisition, and later cohorts have shorter follow-up periods. The resulting segmentation and cohort findings are presented in Section 4.

4 Customer Segmentation and Cohort Analysis

4.1 RFM Segment Profiles

The RFM procedure described in Section 3.5 divides the 5,281 customers in the final evaluation population into four behavioral segments. Table 3 presents their distribution and observed repeat-purchase rates over the subsequent 90-day period. The largest segment is Lost (29.79%), followed by Champions (28.54%), At-risk (21.51%), and Loyal (20.17%). These classifications reflect relative historical purchasing behavior rather than verified customer churn or loyalty status.

Table 3: RFM segment distribution and observed 90-day repeat-purchase rates at the final evaluation snapshot ($N = 5{,}281$).

Segment	Customers	Share (%)	Repeat rate (%)
Champions	1,507	28.54	76.51
Loyal	1,065	20.17	46.57
At-risk	1,136	21.51	33.01
Lost	1,573	29.79	16.91
Total	5,281	100.00	43.36

Repeat-purchase rates vary substantially across segments, from 76.51% among Champions to 16.91% among Lost customers. This ordering indicates that RFM scores capture meaningful differences in subsequent purchasing behavior, supporting their use for descriptive customer profiling and as an interpretable targeting benchmark. However, a higher likelihood of purchasing again does not necessarily justify a discount: customers who would purchase without an incentive may generate lower margins when promoted. The financial value of contacting each customer must therefore be evaluated separately from historical engagement and observed repeat purchasing.

4.2 Cohort Purchasing Activity

Cohort analysis complements RFM segmentation by examining purchasing activity over time. Customers are grouped by the month of their first observed valid purchase, and monthly activity is calculated as the proportion of the original cohort making at least one valid purchase in a given subsequent month. The analysis covers 22 first-observed purchase cohorts, from December 2009 to September 2011. Figure 3 presents activity rates across completed observation months.

The heatmap reveals variation in purchasing activity across cohorts and observation periods. For example, among the 955 customers in the December 2009 cohort, 35.29% purchased in the following month, while 37.59% were active in cohort month 12. This illustrates that monthly activity need not decrease monotonically, since customers may resume purchasing after periods of inactivity. These values describe the proportion purchasing in each month, not continuous retention or customer survival probabilities.

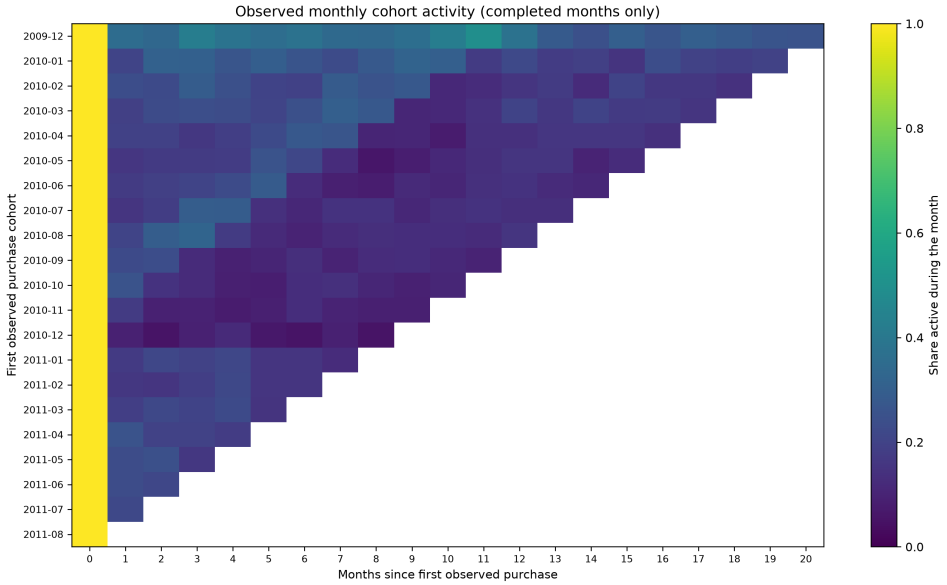


Figure 3: Monthly purchasing activity by first-observed purchase cohort. Month 0 equals 100% by construction; blank cells represent cohort-month combinations without complete observation.

Cohort comparisons are subject to two limitations. First, the earliest observed purchase may not represent a customer's actual first purchase because transactions before December 2009 are unavailable. Second, newer cohorts have shorter observable histories, and the incomplete final observation month is excluded from the heatmap. Consequently, the results provide descriptive evidence of recurring purchasing patterns rather than estimates of acquisition effectiveness or causal retention.

4.3 Implications for Customer Targeting

Together, RFM segmentation and cohort analysis demonstrate substantial heterogeneity in customer purchasing behavior. RFM provides an interpretable summary of customer engagement, while cohort analysis reveals patterns of recurring activity over time. These findings can inform customer monitoring and retention planning, but neither approach identifies the incremental effect of a promotional offer. Accordingly, the subsequent promotion analysis uses customer-level spending forecasts and explicit economic assumptions to evaluate whether discounted contacts generate positive simulated net contribution.

5 Predictive Modeling and Experimental Setup

5.1 Predictive Models

The predictive analysis addresses two supervised learning tasks: estimating the probability of a repeat purchase and forecasting positive gross sales over the subsequent 90 days. Both tasks use the seven historical behavioral features defined in Table 2. The customer–decision-date unit, outcome definitions, and feature construction procedures are described in Section 3. Customer identifiers, country, RFM segments, and future outcomes are excluded from the predictive feature vector.

Logistic Regression and Linear Regression serve as interpretable baselines for classification and regression, respectively. The Logistic Regression model uses a `StandardScaler` fitted exclusively on the training partition, followed by `LogisticRegression` with `max_iter=1000` and `random_state=42`. Linear Regression operates on the original numerical feature scales with the estimator's default configuration. Because linear predictions may be negative, future-sales forecasts are clipped at zero before evaluation and subsequent economic calculations. For comparison, naive predictors assign the training-set repeat-purchase prevalence or mean future gross sales to every evaluation customer.

Gradient Boosting Classifier and Gradient Boosting Regressor are evaluated as nonlinear alternatives using the same seven predictors. Both employ scikit-learn implementations with 100 estimators, a learning rate of 0.1, maximum tree depth of 3, and random seed 42. Other hyperparameters follow the pinned library defaults. No extensive hyperparameter search is conducted; one configured Gradient Boosting candidate is evaluated per task. The purpose is to assess whether additional nonlinear modeling capacity improves predictive performance relative to the interpretable baselines.

5.2 Temporal Training and Validation

To approximate real business decisions and prevent future information from entering historical predictions, the study uses chronological training, validation, and final-test snapshots instead of a random train–test split. Historical observation windows expand from 1 December 2009, while each prediction targets the subsequent 90-day interval. Table 4 summarizes the primary evaluation schedule.

Table 4: Primary temporal evaluation schedule. Outcome end dates are exclusive.

Role	Decision date	Customers	Outcome end
Training 1	2010-12-01	4,266	2011-03-01
Training 2	2011-03-01	4,537	2011-05-30
Validation	2011-06-01	4,933	2011-08-30
Final test	2011-09-10	5,281	2011-12-09

For validation, models are fitted on the first two training snapshots, comprising 8,803 customer–decision-date observations, and evaluated on 4,933 validation observations. Final-test models are subsequently refitted using the two training snapshots and validation data, totaling 13,736 fitting observations, before evaluation on 5,281 customers. These counts refer to snapshots rather than distinct individuals across periods, since customers may appear at multiple decision dates.

Temporal validity is enforced through three safeguards. First, behavioral features use only transactions recorded strictly before the associated decision date. Second, fitting outcomes must be fully observable before the evaluation date; the latest training outcome ends on 30 May 2011 for validation and 30 August 2011 for final testing. Third, preprocessing parameters and model-selection decisions are determined without using final-test information. The final-test predictions are also validated against their expected temporal provenance and prediction source.

The downstream promotion analysis uses the predefined baseline future-sales forecasts, regardless of whether an advanced model satisfies the validation selection criteria. This separation prevents retrospective changes to the business prediction source based on final-test performance.

5.3 Evaluation Metrics and Model Selection

Classification performance is evaluated primarily using Average Precision (AP), reported as PR-AUC in the project artifacts, together with ROC-AUC. Average Precision is computed using scikit-learn's `average_precision_score`, rather than trapezoidal integration of the precision–recall curve. Probability calibration is assessed through the Brier score and Expected Calibration Error (ECE), calculated using ten equal-width probability bins. Lower Brier scores and ECE values indicate better probabilistic agreement with observed outcomes, although ECE depends on the binning procedure.

Future gross-sales regression is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), expressed in GBP per customer over the 90-day horizon. MAE summarizes the average magnitude of errors, while RMSE assigns greater weight to large deviations. Median absolute error and the 90th percentile of absolute error provide supplementary diagnostics for comparing regression models. All regression metrics are calculated after clipping negative forecasts to zero.

Advanced-model adoption follows predefined validation-only criteria. The Gradient Boosting classifier must improve AP by at least 0.01 without increasing the Brier score by more than 0.005. The Gradient Boosting regressor must reduce MAE by at least 5% without increasing the 90th-percentile absolute error by more than 5%. These thresholds are deterministic model-selection rules rather than statistical significance tests. Final-test outcomes are used exclusively to evaluate the resulting models and assess temporal generalization; the actual comparative results are presented in Section 6.

5.4 Calibration, Interpretation, and Robustness Procedures

The study additionally evaluates sigmoid probability calibration for Logistic Regression. The calibration procedure fits a logistic mapping to the base classifier's decision scores using chronologically separated data. For validation, the base classifier is fitted on Training 1 and the calibrator on Training 2; for final testing, the base classifier uses both training snapshots and the calibrator uses validation data. Because this allocation differs from the ordinary uncalibrated fitting procedure, metric differences cannot be attributed solely to the calibration transformation. Calibration is retained only when validation Brier score improves by at least 0.002 without reducing AP by more than 0.005.

To examine model interpretability, Tree SHAP is applied to the final Gradient Boosting classifier using a deterministic sample of 750 final-test customers. Global feature importance is summarized through mean absolute SHAP attributions. These values describe contributions within the predictive model and are not interpreted as causal effects or direct percentage-point changes in purchasing probability.

Uncertainty in baseline evaluation metrics is assessed using 1,000 seeded bootstrap resamples of evaluation customers, with fitted predictions held fixed. The 2.5th and 97.5th percentiles provide empirical intervals conditional on the fitted models; they do not capture model-refitting uncertainty, future temporal shifts, or promotional treatment effects. A separate rolling-origin evaluation uses five historical evaluation origins to investigate temporal performance stability, with results reported in Section 8.

Reproducibility is supported by a pinned Python environment, fixed random seeds, centralized configuration, automated tests, and stored model-evaluation artifacts. Detailed implementation settings and analytical provenance are maintained in the accompanying repository and model documentation.

6 Model Evaluation and Interpretation

6.1 Predictive Performance

The predictive models are evaluated on the chronological validation and final-test snapshots described in Section 5. The evaluation examines whether historical behavioral features improve forecasting relative to naive predictors and whether Gradient Boosting provides additional predictive value over interpretable baselines. All results are calculated using out-of-time customer observations, with final-test outcomes excluded from model fitting and selection.

Table 5 reports classification performance. Logistic Regression substantially outperforms the constant-probability benchmark, achieving an Average Precision (AP) of 0.6953 on validation and 0.7502 on the final test. Gradient Boosting improves AP to 0.7189 and 0.7691, respectively, with modest improvements in ROC-AUC. These results indicate that nonlinear modeling provides additional ranking capability for identifying customers likely to make a repeat purchase.

Table 5: Repeat-purchase classification performance under chronological evaluation.

Split	Model	AP	ROC-AUC	Brier
Validation	Naive	0.3286	–	–
	Logistic	0.6953	0.8060	0.1614
	GBM	0.7189	0.8150	0.1559
Final test	Naive	0.4336	–	–
	Logistic	0.7502	0.7900	0.1995
	GBM	0.7691	0.7987	0.1994

The increase in AP between validation and final testing should be interpreted alongside the increase in observed repeat-purchase prevalence, from 32.86% to 43.36%. Since AP depends on class prevalence, its increase does not necessarily indicate improved discrimination over time. Indeed, Logistic Regression's ROC-AUC decreases from 0.8060 to 0.7900. The results nevertheless support the usefulness of historical purchasing behavior for customer ranking, with Gradient Boosting providing a modest advantage on the evaluated periods.

Regression results are summarized in Table 6. Linear Regression reduces final-test MAE from £636.34 for the naive benchmark to £395.69, while RMSE decreases from £3,288.87 to £2,016.14. Gradient Boosting achieves a lower validation MAE (£277.66 versus £294.53), but this improvement does not generalize to the final test, where its MAE rises to £400.40 compared with £395.69 for Linear Regression. Gradient Boosting also produces higher RMSE in both periods.

Table 6: Future gross-sales regression performance (GBP per customer over 90 days).

Split	Model	MAE	RMSE
Validation	Naive	513.04	2,057.12
	Linear	294.53	1,030.92
	GBM	277.66	1,120.81
Final test	Naive	636.34	3,288.87
	Linear	395.69	2,016.14
	GBM	400.40	2,167.29

The substantially higher RMSE relative to MAE reflects the difficulty of forecasting the highly skewed distribution of customer spending. On the final test, observed future gross sales average £543.11 per customer, compared with a mean baseline prediction of £379.00. Moreover, 1,583 negative raw Linear Regression forecasts are clipped to zero. These findings indicate that the baseline provides useful monetary predictions but remains sensitive to customers with unusually large spending outcomes.

Both Gradient Boosting candidates satisfy the predefined validation selection criteria. However, the regression reversal on the final test demonstrates that greater model complexity does not guarantee better temporal generalization. Accordingly, the downstream promotion analysis retains the predefined Linear Regression forecasts rather than switching prediction sources retrospectively.

6.2 Probability Calibration and Model Interpretation

Classification ranking performance does not necessarily imply accurate probability estimates. To examine probabilistic reliability, the study evaluates Brier scores, Expected Calibration Error (ECE), and reliability curves. Figure 4 compares predicted repeat-purchase probabilities with observed purchasing frequencies on the final temporal test.

Both principal classifiers exhibit substantially worse calibration on the final test. Logistic Regression's ECE increases from 0.0261 on validation to 0.1170 on the final test, while Gradient Boosting's ECE increases from 0.0181 to 0.1301. Although Gradient Boosting achieves higher final-test AP, its calibration error is larger than that of Logistic Regression. These results emphasize that stronger ranking performance does not guarantee more reliable absolute purchase probabilities.

The sigmoid-calibrated Logistic Regression candidate reduces validation Brier score only from 0.161353 to 0.161257, an improvement of approximately 0.000096. This falls below the predefined adoption threshold of 0.002, so the calibration candidate is not retained. Its slightly lower final-test Brier score cannot override the validation-based selection rule. Moreover, the calibration experiment uses different chronological fitting allocations from the uncalibrated comparator, meaning the observed metric differences cannot be attributed exclusively to the sigmoid transformation.

Model interpretation is examined using Tree SHAP applied to the Gradient Boosting classifier. Figure 5 presents the feature-attribution distribution for a deterministic sample of 750 final-test customers. Purchase frequency and recency exhibit the largest mean absolute SHAP attributions, followed by historical monetary value and tenure. This indicates that the fitted classifier relies primarily on how frequently and recently customers have purchased, with additional information provided by spending-related characteristics.

SHAP values describe contributions within the fitted predictive model and should not be interpreted as independent causal effects. Correlated behavioral features may share predictive information, and the attribution magnitudes are not established as direct changes in purchase probability. Thus, the analysis provides insight into the model's predictive behavior without identifying which customer characteristics could be manipulated to increase purchasing.

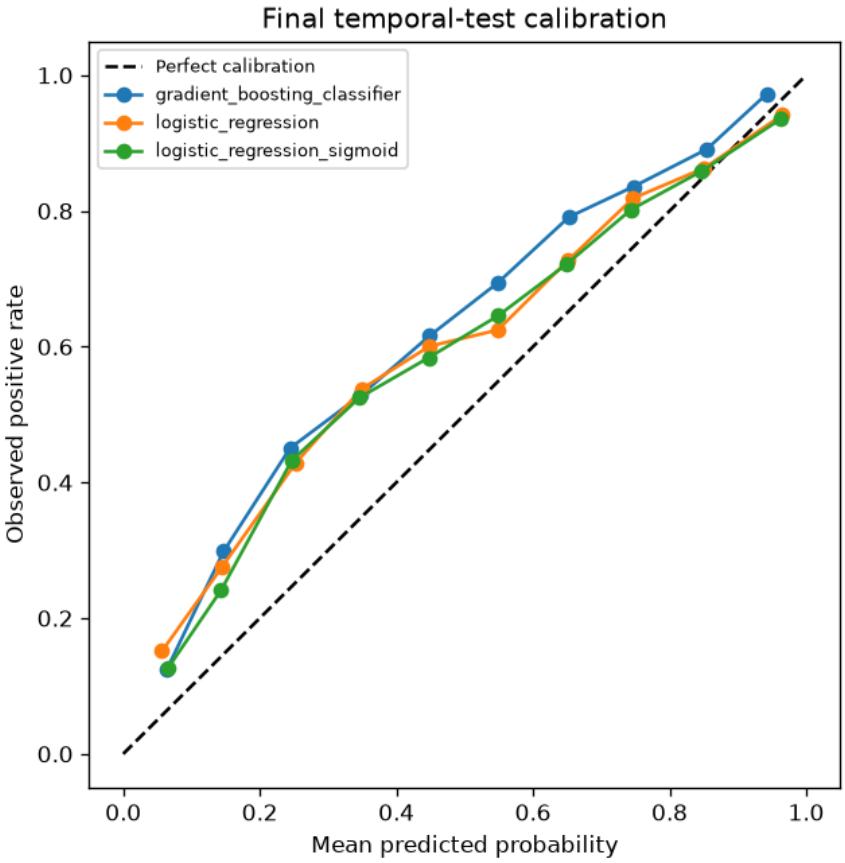


Figure 4: Reliability diagram of 90-day repeat-purchase predictions on the final temporal test ($N = 5{,}281$). The dashed line represents perfect calibration. The sigmoid-calibrated Logistic Regression is shown for comparison but was not retained under the validation-based selection criterion.

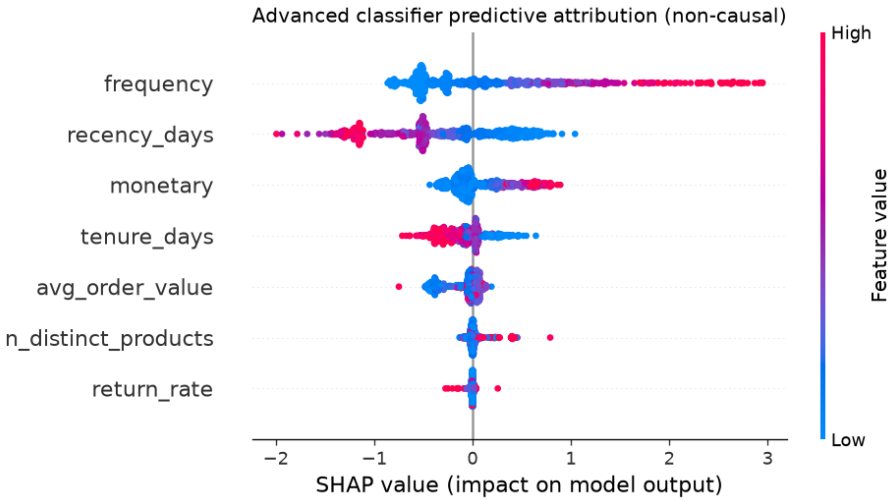


Figure 5: Tree SHAP summary for the Gradient Boosting repeat-purchase classifier, based on 750 final-test customers. Attributions describe predictive relationships, not causal effects.

6.3 Uncertainty and Business Implications

Baseline metric uncertainty is assessed through 1,000 customer-level bootstrap resamples, keeping fitted predictions fixed. On the final test, the empirical 95% percentile intervals are [0.7312, 0.7688] for AP and [0.7776, 0.8020] for ROC-AUC. The corresponding regression intervals are [£350.89, £453.53] for MAE and [£1,188.64, £2,805.03] for RMSE. The wider monetary-error intervals, particularly for RMSE, indicate substantial sensitivity to high-spending customers. These intervals describe evaluation-sample variability conditional on the fitted models; they do not account for model-refitting uncertainty, future distribution shifts, or uncertainty in promotional responses.

Overall, the results support the use of historical transaction features for predicting repeat purchases and future gross sales, while revealing limitations in monetary forecasting and probability calibration. Gradient Boosting improves classification ranking but provides inconsistent regression gains across temporal periods. Importantly, predictive accuracy alone does not establish whether a customer should receive a discount. A high predicted purchase probability or future-sales value may still correspond to a negative promotional contribution once discount-related margin erosion and contact costs are considered. Section 7 therefore evaluates customer-level forecasts through an explicit economic decision model, while additional temporal robustness evidence is presented in Section 8.

7 Promotion Targeting and Business Impact

7.1 Promotion Economics and Scenario Assumptions

The predictive models estimate future purchasing behavior, but a high predicted customer value does not necessarily justify offering a discount. A promotion is economically beneficial only when the additional contribution generated under the assumed customer response exceeds discount-related margin erosion and contact costs. The decision framework therefore converts customer-level gross-sales forecasts into simulated net incremental margins.

Let $v_i \geq 0$ denote the baseline model's nonnegative forecast of customer i's 90-day gross sales, g the gross-margin rate, d_s the discount rate in scenario s, and c the cost of contacting one customer. The assumed proportional increase in sales volume is $b_s m_{z(i)}$, where b_s is the scenario's base growth assumption and $m_{z(i)}$ is the multiplier associated with customer i's RFM segment. Assuming unchanged unit product costs, the simulated net incremental margin from contacting customer i is

$$\Delta M_{i,s} = v_i \left[(1 + b_s m_{z(i)})(g - d_s) - g\right] - c. \tag{1}$$

The expression compares the promoted gross margin, after accounting for the assumed increase in sales volume and the price discount, with the gross margin that would be associated with the reference sales forecast. Contact cost is deducted separately. The regression forecast is unconditional on repeat purchasing and is therefore not multiplied by the predicted repeat-purchase probability. Neither the baseline forecast nor the assumed sales-volume response is an experimentally identified counterfactual outcome.

Three promotional scenarios are evaluated using a common gross-margin assumption of $g = 0.40$ and contact cost of $c = £0.50$. The segment multipliers are 0.5 for Champions, 1.0 for Loyal, 1.3 for At-risk, and 0.4 for Lost customers. Table 7 summarizes the discount and growth assumptions. The Moderate scenario serves as the reference configuration for saved customer-level recommendations.

Table 7: Configured promotion scenarios and maximum pre-contact incremental-margin coefficients.

Scenario	Discount	Base growth	Max. coefficient
Conservative	5%	10%	−0.0045
Moderate	10%	20%	−0.0220
Aggressive	20%	35%	−0.1090

All growth rates and segment multipliers are planning assumptions rather than measured promotional treatment effects. The maximum coefficient in the final column is the largest value of $(1 + b_s m_z)(g - d_s) - g$ across the four RFM segments. Its economic implications are examined alongside the policy results below.

7.2 Budget-Constrained Targeting and Policy Benchmarks

The retailer is assumed to have a direct-contact budget of $B = £750$. At a uniform cost of $£0.50$ per contact, the maximum campaign capacity is 1,500 customers from the final evaluation population of 5,281. This budget constrains contact expenditure; discount-related margin erosion is already incorporated into Equation 1.

Let $x_i \in \{0, 1\}$ indicate whether customer i receives a promotional contact. For each scenario, the proposed policy solves

$$\begin{aligned} \max_{\boldsymbol{x}} \quad & \sum_{i=1}^{N} x_i \Delta M_{i,s} \\ \text{subject to} \quad & c \sum_{i=1}^{N} x_i \leq B, \\ & x_i \in \{0, 1\}, \quad i = 1, \ldots, N. \end{aligned} \tag{2}$$

Because contact costs are uniform and customer contributions are additive, the implemented solution ranks customers by simulated net incremental margin and selects up to 1,500 customers with strictly positive contributions. The budget need not be exhausted, and selecting no customers is feasible. This ranking rule is optimal for the stated uniform-cost formulation, which does not model customer interactions, heterogeneous costs, or competing offers.

Four policies are compared under the same maximum contact capacity. *No Promotion* contacts nobody and provides a zero incremental-margin reference. *Non-targeted Promotion* randomly selects 1,500 customers without replacement, with results averaged across 1,000 seeded simulations. *RFM-based Targeting* selects the 1,500 highest-ranked customers according to RFM score and historical monetary value. The *Proposed Policy* selects only customers with positive simulated net incremental margins, up to the budget limit.

Unlike Non-targeted and RFM-based Targeting, the Proposed Policy may leave budget unused. The comparison therefore uses an equal maximum capacity, not necessarily equal realized contact expenditure. All policies are evaluated on the same final-test customer population using the predefined baseline future-sales forecasts and identical economic assumptions within each scenario.

7.3 Simulated Promotion Outcomes

Table 8 reports the simulated net incremental margins of the four policies across the three scenarios. The Non-targeted results are averages over 1,000 random campaign selections; all monetary values represent changes relative to No Promotion rather than total retailer profit.

Table 8: Simulated net incremental margin (GBP) across promotion policies.

Scenario	No promo.	Non-targeted	RFM	Proposed
Conservative	0	−17,417.59	−55,333.45	0
Moderate	0	−37,407.57	−118,314.35	0
Aggressive	0	−90,677.15	−277,865.97	0

Both Non-targeted and RFM-based Targeting produce negative simulated incremental margins in every reference scenario. The losses become larger as the configured discounts increase. RFM-based Targeting performs particularly poorly because prioritizing customers with substantial historical and predicted spending can magnify discount-related margin erosion when the assumed additional sales volume is insufficient. Both conventional policies contact 1,500 customers and incur the full $£750$ contact cost in each scenario.

In contrast, the Proposed Policy selects no customers and produces zero incremental margin and contact expenditure across all three scenarios. This result follows directly from the economic assumptions rather than from an unexpectedly restrictive customer-ranking procedure. Defining the pre-contact contribution coefficient as

$$\alpha_{z,s} = (1 + b_s m_z)(g - d_s) - g, \tag{3}$$

Equation 1 becomes $\Delta M_{i,s} = v_i \alpha_{z(i),s} - c$. The At-risk segment has the highest assumed growth multiplier and therefore the largest coefficient within each scenario. Nevertheless, the maximum coefficients are -0.0045, -0.0220, and -0.1090 for the Conservative, Moderate, and Aggressive scenarios, respectively, as shown in Table 7.

Since every coefficient is negative, $v_i \geq 0$, and $c > 0$, all customer-level simulated contributions are strictly negative. Consequently, no customer satisfies the positive-contribution selection rule, and the optimal solution to Equation 2 is to contact nobody. This conclusion is structural under the configured assumptions: it does not demonstrate that the predictive model has identified customers who would be unresponsive to promotional offers.

The Non-targeted Monte Carlo results also remain negative across the simulated customer selections. For example, the Moderate scenario produces a mean incremental margin of $-£37{,}407.57$, with 2.5th and 97.5th percentiles of $-£46{,}583.42$ and $-£30{,}081.21$. These percentiles describe variation between random customer lists conditional on the same forecasts and assumptions; they are not confidence intervals for causal campaign effects.

7.4 Managerial Implications

The results indicate that promotional targeting should be evaluated against incremental economic contribution rather than campaign reach or historical customer value alone. Under the three configured scenarios, spending the available contact budget would reduce simulated margin, while abstaining preserves the No Promotion reference. The difference between the Proposed and Non-targeted policies therefore represents an avoided *simulated* loss, not a realized financial saving.

The customer-level decision output also distinguishes promotional eligibility from broader CRM recommendations. Under the saved Moderate scenario, 2,454 customers receive a *Retain* label, 2,827 receive *No-offer*, and none qualify for *Nurture*. The Retain label identifies positive-value Champions or Loyal customers for general relationship management; it does not authorize a discounted contact. Actual campaign selection is determined separately by the positive-margin optimization rule, which selects no customers under the reference assumptions.

The principal recommendation is therefore not that promotions are universally unprofitable, but that the three configured campaigns should not be launched on the basis of their assumed economics. The profitability of alternative offers depends on the relationship between discount depth, additional sales volume, customer value, and costs. Section 8 examines the corresponding break-even conditions and identifies alternative assumptions under which promotional contacts become economically feasible. Since the source data do not identify promotional treatment effects, these results support conditional scenario planning rather than verified campaign effectiveness.

8 Sensitivity and Robustness Analysis

8.1 Break-Even Analysis

The promotion results in Section 7 indicate that all three reference scenarios favor no promotional contact. Since this conclusion depends on configured economic assumptions rather than measured treatment effects, break-even analysis is used to identify the conditions under which a promotion could generate positive simulated incremental margin.

From Equation 1, setting the customer-level contribution to zero and solving for the required base sales-growth rate gives

$$b_i^* = \frac{1}{m_{z(i)}} \left(\frac{g + c/v_i}{g - d} - 1 \right), \tag{4}$$

where $v_i > 0$, $g > d$, and $m_{z(i)} > 0$. Alternatively, the maximum economically supportable discount for a given growth assumption is

$$d_i^* = g - \frac{g + c/v_i}{1 + b\, m_{z(i)}}. \tag{5}$$

These expressions describe financial thresholds within the scenario model, not estimates of actual customer responses. Customers with zero predicted future sales have no defined break-even growth threshold and cannot cover a positive contact cost. Furthermore, reaching exact break-even produces zero incremental margin, whereas the proposed policy requires a strictly positive contribution.

Table 9 summarizes the median thresholds for customers with positive predicted future sales under the Moderate scenario, with a 10% discount and 20% assumed base growth.

Table 9: Median break-even requirements by RFM segment under the Moderate scenario.

Segment	**Customers**	**Required growth (%)**	**Max. discount (%)**
Champions	1,503	67.38	3.54
Loyal	951	34.47	6.38
At-risk	779	26.75	7.91
Lost	465	90.49	2.17

At-risk customers have the lowest median required base growth, approximately 26.75%, reflecting their relatively favorable assumed response multiplier. Nevertheless, this threshold exceeds the configured 20% growth assumption. Their median supportable discount is also only 7.91%, below the reference discount of 10%. Other segments require even greater assumed growth or lower discounts. These results explain why the Moderate scenario is economically unfavorable and provide quantitative thresholds against which alternative campaign proposals can be assessed.

8.2 Economic Sensitivity Analysis

Economic sensitivity is examined by varying the assumed discount rate, base sales growth, gross margin, and segment response multipliers around the Moderate scenario. For each parameter configuration, customer-level incremental contributions are recalculated and the budget-constrained policy reselects eligible customers. Thus, changes in simulated campaign margin reflect both altered financial contributions and changes in the selected customer population.

In the one-at-a-time analysis, reducing the discount from 10% to 5% while maintaining 20% assumed base growth produces a simulated net incremental margin of £7,175.92. Increasing assumed base growth to 35% at the reference 10% discount produces £3,990.80. By comparison, increasing the assumed gross margin from 40% to 50% yields £143.77, while increasing the At-risk response multiplier from 1.3 to 1.69 yields £15.05. These results identify discount depth and assumed growth as influential parameters within the tested ranges, although the different parameter ranges prevent interpreting the changes as directly comparable economic elasticities.

To examine the interaction between discount and growth assumptions, the study evaluates a two-dimensional grid with discounts of 5%, 10%, 15%, and 20%, and assumed base growth ranging from 10% to 50%. Other parameters, including the 40% gross margin, £0.50 contact cost, £750 budget, and segment multipliers, remain fixed. Figure 6 presents the resulting simulated net incremental margins.

Positive simulated margins emerge when the assumed additional sales volume is sufficiently large relative to the discount. At a 5% discount and 20% base growth, the policy selects 1,500 customers and generates £7,175.92 in simulated net margin. At a 10% discount and 30% growth, 664 customers qualify, producing £1,424.27 with £332 in contact expenditure. At a 15% discount and 50% growth, 616 customers qualify, yielding £962.39. No positive-margin campaign is identified at a 20% discount within the evaluated growth range.

These combinations represent the first profitable points in the tested discrete grid, not exact continuous break-even thresholds. More importantly, positive simulated contributions do not establish that the assumed sales responses are achievable. The analysis demonstrates how the targeting policy adapts to alternative economics, while emphasizing that actual promotion effectiveness must be established separately.

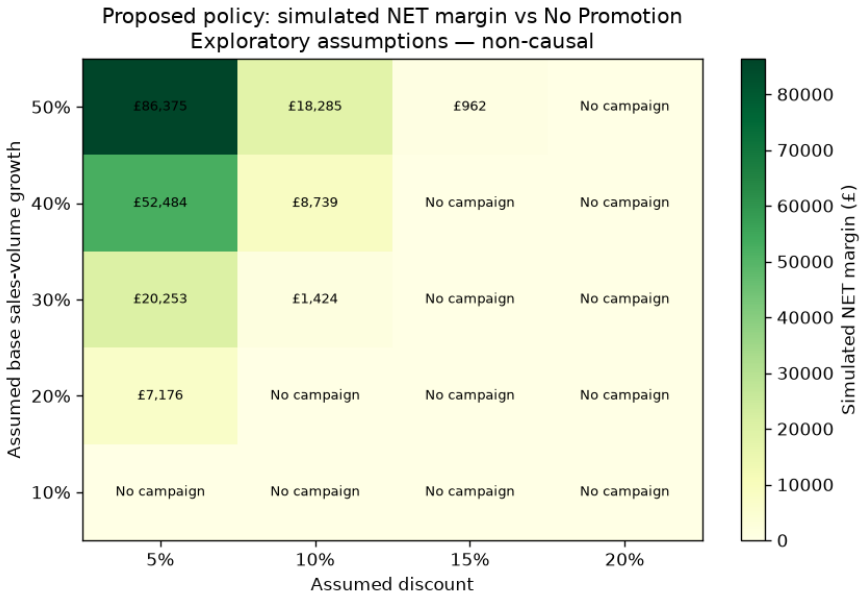


Figure 6: Simulated net incremental margin across discount and assumed sales-growth combinations. Zero values indicate that the proposed policy selects no customers.

8.3 Temporal Robustness

Predictive performance may vary across decision dates because customer activity, purchasing prevalence, and spending distributions change over time. To assess this sensitivity, the study conducts a secondary rolling-origin evaluation over five historical decision dates. At each origin, models are fitted using eligible earlier snapshots with fully observed outcome windows. This design supplements the main temporal evaluation rather than replacing it.

Table 10 compares Average Precision (AP) for repeat-purchase classification and Mean Absolute Error (MAE) for gross-sales regression across the five origins.

Table 10: Secondary rolling-origin predictive performance. Regression MAE is in GBP.

	Classification AP		**Regression MAE**	
Decision date	**Logistic**	**GBM**	**Linear**	**GBM**
2010-09-01	0.7879	0.7897	556.91	525.66
2010-12-01	0.6598	0.6617	609.32	617.95
2011-03-01	0.6799	0.6850	690.04	716.31
2011-06-01	0.7061	0.7180	335.03	302.21
2011-09-10	0.7560	0.7664	397.64	408.16

Gradient Boosting achieves higher classification AP at all five origins, although the improvements are generally modest. Its regression performance is less consistent: MAE is lower than the Linear Regression baseline at only two of the five origins. These findings reinforce the distinction observed in the main evaluation, where the nonlinear classifier improves customer ranking but the advanced regressor does not consistently outperform the baseline.

The final rolling-origin evaluation shares the decision date of 10 September 2011 with the primary final test, but uses 19,760 fitting observations rather than 13,736. Consequently, the corresponding metrics differ slightly and should not be interpreted as repeated evaluations of an identical fitted model. The rolling-origin results provide evidence about variation across historical periods, not a guarantee of future predictive stability.

8.4 Data-Preparation Robustness

The primary analysis retains exact duplicate transaction lines because their underlying semantics cannot be established from the available identifiers. To assess the influence of this decision, a separate robustness experiment removes 34,335 exact duplicate rows, reducing the dataset from 1,067,371 to 1,033,036 records while preserving the main temporal and modeling specifications.

Duplicate removal leaves the final evaluation population unchanged at 5,281 customers, with the same 43.36% repeat-purchase rate. Classification AP changes negligibly from 0.750184 to 0.750191, while regression MAE increases from £395.69 to £396.54. RMSE increases from £2,016.14 to £2,066.55, indicating somewhat greater sensitivity among larger monetary prediction errors. Overall, the predictive conclusions remain broadly similar, although the experiment does not establish which duplicate-handling policy is objectively correct.

The proposed promotion policy continues to select no customers under all three reference scenarios after duplicate removal. However, this stability follows primarily from the negative incremental-margin coefficients established in Section 7. Because those coefficients remain negative for every segment, the policy abstains for any nonnegative future-sales forecast under the same economic assumptions. The unchanged decision should therefore not be interpreted as independent evidence of forecasting robustness.

Taken together, the analyses indicate that the no-promotion recommendation is sensitive to economic assumptions but unchanged under the tested duplicate-handling alternative. Predictive performance also varies across historical periods, particularly for monetary regression. The practical implication is that future campaigns should be evaluated against explicit break-even conditions and supported by measured promotional response evidence before deployment.

9 Dashboard and Managerial Implications

9.1 Dashboard Implementation and Data Scope

An interactive decision-support dashboard was developed using Streamlit to translate the analytical results into interpretable customer and promotion insights. The application integrates the final-test customer score table, RFM segmentation, cohort activity, predictive outputs, and precomputed promotion-policy comparisons. Before displaying the results, it validates the temporal provenance and prediction source of the customer data. The dashboard operates on the historical evaluation population of 5,281 customers at the decision date of 10 September 2011, rather than on a live customer database. Figure 7 illustrates an exploratory scenario in which the discount rate is reduced to 5% while assumed sales growth remains at 20%. The dashboard identifies economically eligible customers and allocates the available contact budget according to simulated net incremental contribution.

The application separates predictive modeling from interactive economic analysis. Customer-level predictions are generated by the analytical pipeline before the dashboard is launched. When users modify promotional assumptions, the dashboard recalculates simulated incremental contributions and targeting decisions using these existing forecasts, without retraining the models or overwriting the original analytical artifacts. This design enables interactive scenario exploration while keeping the predictive inputs consistent.

Two analytical scopes are maintained. The saved global comparison presents the four promotion policies evaluated on the complete final-test population, whereas the interactive scenario analysis recalculates the Proposed Policy within the currently filtered customer subset. The cohort heatmap also represents the full historical population and does not change with customer filters. Consequently, global benchmark results and filtered scenario outcomes must not be interpreted as directly comparable estimates for identical customer populations.

9.2 Interactive Scenario Analysis and Customer Decisions

The dashboard allows users to filter customers by country, RFM segment, and saved reference-scenario recommendation. A Scenario Sandbox provides adjustable discount rates, assumed sales-volume growth, gross margins, and contact budgets. The per-customer contact cost and segment-specific growth multipliers remain fixed at their configured values. For each selected combination, the

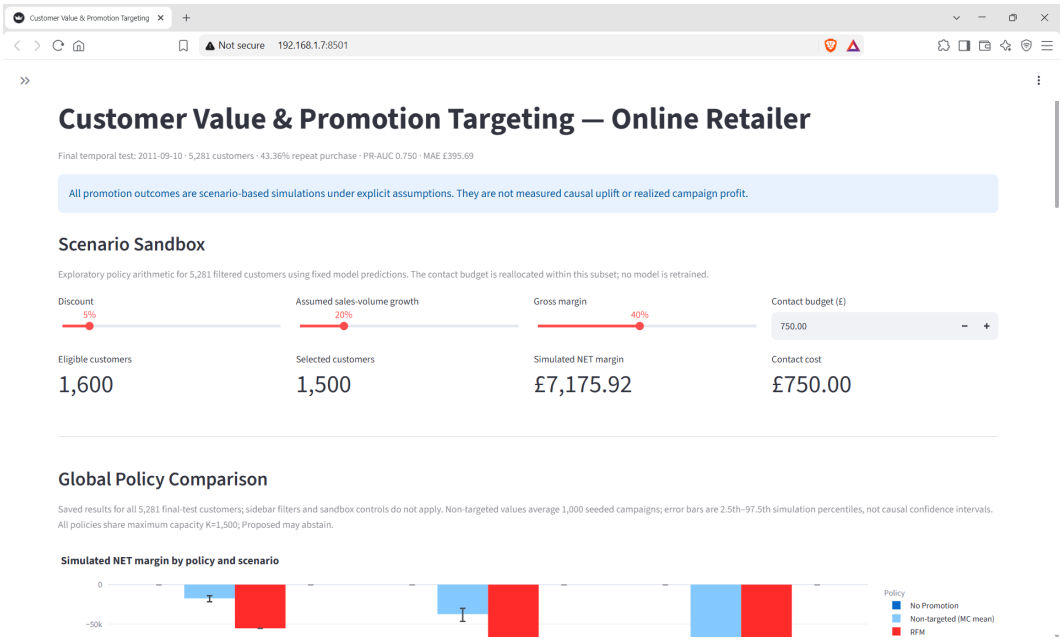


Figure 7: Streamlit Scenario Sandbox under an exploratory promotion scenario (5% discount, 20% assumed sales growth, and a £750 contact budget). The policy selects 1,500 customers and generates £7,175.92 in simulated net incremental margin. Results are assumption-based, not causally estimated campaign effects.

application recalculates customer-level simulated net incremental margins and applies the budget-constrained targeting rule introduced in Section 7. The recommendation filter refers to the saved reference scenario and is not automatically redefined when sandbox assumptions change.

Four decision-oriented KPIs summarize the resulting scenario: economically eligible customers, selected customers, simulated net incremental margin, and total contact expenditure.

The interface also provides customer-segment visualizations, predicted purchasing indicators, and the saved global policy comparison. When the current assumptions produce eligible contacts, the selected customers are ranked by simulated net incremental contribution and can be exported as a CSV file. The export includes customer identifiers, historical attributes, predictions, scenario-specific contributions, and the economic assumptions used to generate the selection. These details help ensure that customer recommendations remain interpretable in the context of the scenario that produced them.

Under the default Moderate scenario, the dashboard reports no eligible promotional contacts, consistent with the results in Section 7. Alternative assumptions can produce positive simulated margins and nonempty customer selections, as demonstrated in Section 8. The dashboard therefore serves as a what-if analysis tool: changing an assumed growth rate alters the simulated decision but does not establish that customers would actually respond to a promotion at that rate.

9.3 Managerial Use and Deployment Considerations

The dashboard supports a decision process in which managers first examine the global policy evidence, then evaluate alternative promotional assumptions and inspect any economically eligible customer groups. Discount levels and growth expectations should be supported by credible business evidence rather than adjusted solely to obtain favorable simulated results. Even when the model identifies positive-margin contacts, managers must consider operational feasibility, customer contact permissions, and constraints that are not represented in the current optimization.

The principal managerial implication is that promotional budgets should be treated as upper limits rather than expenditure targets. Under the configured scenarios, the economically justified decision is

to refrain from discounted contacts, although this does not preclude non-discount retention activities. More generally, the dashboard helps distinguish customers who are valuable or likely to purchase from customers whose promotional contact is financially justified under a specified set of assumptions.

The application remains a retrospective decision-support prototype, not a production CRM or campaign-execution system. Its customer-level exports do not establish that offers were delivered or that incremental profits were realized. Before operational deployment, the retailer would need current customer data, validated financial inputs, appropriate access and privacy controls, and procedures for monitoring predictive performance. A randomized promotional experiment with a suitable treatment and control design would also be required to estimate actual incremental responses and assess whether the scenario-based recommendations translate into realized financial benefits. These safeguards and experiments are future requirements rather than completed components of the present study.

10 Limitations and Conclusion

10.1 Limitations and Future Work

Several limitations should be considered when interpreting the findings. First, the analysis relies on historical transaction data from a single UK retailer between 2009 and 2011, limiting its generalizability to contemporary retail environments. Approximately 22.77% of the original transaction records lack customer identifiers and are excluded from customer-level modeling, potentially introducing selection bias. The treatment of exact duplicate records also remains uncertain, although its influence is examined through sensitivity analysis. Furthermore, the monetary prediction target represents positive gross sales rather than returns-adjusted revenue or customer profit, while customer tenure and cohort membership may be left-censored by the available observation period.

Second, predictive performance remains subject to temporal and statistical uncertainty. Despite chronological validation, the models are evaluated on a limited number of historical periods, and customer spending exhibits a highly skewed distribution with many zero outcomes. Regression errors are particularly sensitive to high-value customers, while probability calibration deteriorates on the final temporal test. The reported bootstrap intervals also exclude uncertainty from model refitting and future distribution shifts. Consequently, the predictions should be treated as estimates with limitations rather than precise forecasts of individual customer behavior.

The most consequential limitation concerns promotional effectiveness. The dataset does not provide identified treatment and control groups or reliable records of promotional exposure. Therefore, the assumed sales-growth rates and segment multipliers cannot be interpreted as estimated causal responses. The financial model also simplifies real campaign economics through uniform contact costs, a common gross-margin rate, and additive customer contributions, without explicitly accounting for inventory constraints, customer interactions, or heterogeneous campaign expenses. Moreover, the policy benchmarks share a maximum contact capacity but not necessarily equal expenditure. The conclusion that no promotion should be offered under the three reference scenarios follows structurally from their negative incremental-margin coefficients; it does not establish that the predictive model has identified customers with inferior causal promotion responses.

Future research should prioritize causal validation through randomized promotional experiments with appropriately defined treatment and control groups. Such experiments would enable the measurement of realized incremental net contribution and provide evidence for estimating heterogeneous treatment effects through uplift modeling. Additional improvements could include two-stage or distribution-aware models for zero-heavy customer spending, customer-specific promotion costs, and more realistic operational constraints. Validation on recent transactional data, together with ongoing monitoring of model performance and calibration, would also be necessary before deployment.

Finally, the dashboard remains a retrospective decision-support prototype rather than a production campaign system. Its customer-level outputs require appropriate access controls, data minimization, and human oversight before operational use. Although protected characteristics are not directly included among the predictive features, the study does not establish comprehensive fairness or disparate-impact assessments. Privacy, fairness, and governance safeguards should therefore form part of any future implementation rather than being assumed to have been validated.

10.2 Conclusion

This study developed an integrated Business Analytics framework for customer value assessment and promotion targeting using the UCI Online Retail II dataset. By combining behavioral segmentation, 90-day purchasing predictions, and budget-constrained economic optimization, the framework connects historical customer information with explicit marketing decisions. The predictive experiments demonstrate that transaction-based features provide useful information about future purchasing behavior, although nonlinear models do not consistently outperform interpretable baselines across all tasks and temporal periods.

The principal business finding is that none of the three configured promotion scenarios generates positive simulated incremental margin for any customer. Consequently, the proposed policy selects no promotional contacts, avoiding the negative simulated contributions produced by conventional Non-targeted and RFM-based strategies. Sensitivity analysis further demonstrates that alternative discount and assumed-growth combinations can support positive-margin decisions, indicating that the recommendation is conditional rather than universal.

The study's main contribution is therefore not simply the prediction of customer value, but the integration of predictive analytics with transparent economic decision rules that permit financially justified abstention. The results emphasize that valuable customers are not necessarily profitable promotion targets and that marketing expenditure should be guided by incremental contribution rather than campaign reach alone. Nevertheless, because promotional responses are assumed rather than causally estimated, the framework should be used for scenario-based decision support until experimental evidence establishes its effectiveness in practice.

11 Team Member Contributions

The project was completed collaboratively by three team members, with responsibilities distributed across the main analytical components.

Nguyen Quoc Hung (20252568M): Responsible for data acquisition, exploratory data analysis, transaction cleaning, customer-level feature engineering, RFM segmentation, cohort analysis, and duplicate-handling robustness assessment.

Bui Trung Kien (20252567M): Responsible for predictive model development, temporal validation, baseline and Gradient Boosting comparisons, performance evaluation, probability calibration, SHAP interpretation, and rolling-origin robustness analysis.

Nguyen Minh Thuong (20252280M): Responsible for promotion economics, budget-constrained targeting optimization, policy benchmarking, break-even and sensitivity analyses, and development of the interactive Streamlit dashboard.

All members jointly contributed to business problem formulation, methodology discussions, system integration, result interpretation, report writing, and presentation preparation.

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